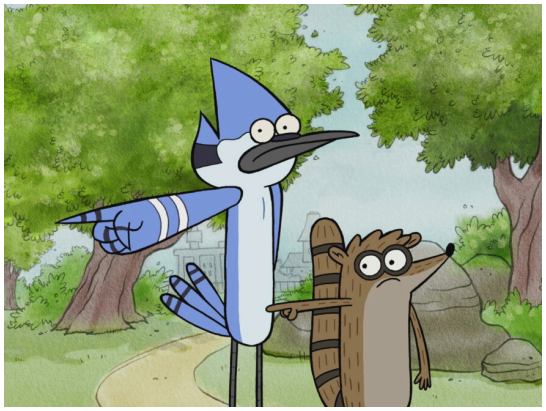
W13 Exercises

📌 Discuss with TAs or consult the TA session notes if you run into any difficulties!



Mordecai (the bird) and Rigby (the racoon)

💬 Mordecai and Rigby are supposed to be working at the park, but they keep getting distracted by playing video games during work hours. Their boss Benson is tired of seeing them wasting time and wants to find out whether their work performance is different, so that they can separate them by assigning them different works

Benson has collected performance scores from Mordecai and Rigby while they are supposed to be working. Higher scores represent better work performance. Complete the following tasks to help Benson find out the answer!

Task 1: Mordecai's Work Performance

- Benson knows that the park's expected average work-performance score is `1200` points per work session
- After catching Mordecai playing video games instead of working, Benson asks him to complete 15 work sessions and records his performance:

```
# Mordecai's performance score
1260, 1190, 1235, 1280, 1215,
1305, 1250, 1205, 1270, 1245,
1290, 1225, 1265, 1185, 1310
```

One-Sample t-test

- Import the performance score data and name it as `mordecai_work`
- Help Benson determine whether Mordecai's mean work performance is **greater than** `1200` points
  - **Hint:** use 95% confidence level
- Based on the p-value, does Benson have sufficient evidence to conclude that Mordecai's average work performance is greater than `1200` ?

Task 2: Mordecai vs Rigby

- Benson also wants to know whether Mordecai and Rigby have different levels of work performance. Rigby claims that he is actually better worker than Mordecai, despite spending lots of time playing video games
- Benson records Rigby's work performance over 12 work sessions:

```
# Rigby's performance score
1320, 1375, 1295, 1340, 1310,
1390, 1355, 1285, 1365, 1330,
1305, 1380
```

Paired Sample t-test

- Import the performance score data and name it as `rigby_work`
- Help Benson determine whether Rigby's average work performance is different from Mordecai's average work performance
  - **Hint:** use two-sided alternative hypothesis and 95% confidence level
- Is Rigby telling the truth? Does he have a different mean work performance from Mordecai?